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Subject : DSA

Unit 3, S8, SLO2 - Queue Implementation Using Array

Enqueue Operation - Algorithm to insert an element NUM in a queue

Step 1: IF REAR = _____

Write OVERFLOW

Goto step 4

[END OF IF]

Step 2: IF FRONT=____ and REAR= _____

SET FRONT = REAR = 0

ELSE

SET REAR = _____

[END OF IF]

Step 3: SET QUEUE[REAR] = _____

Step 4: EXIT

Answer: Step 1: IF REAR = MAX - 1

Step 2: IF FRONT = -1 and REAR = -1

SET FRONT = REAR = 0

ELSE

SET REAR = REAR + 1

Step 3: SET QUEUE[REAR] = NUM

Step 4: EXIT

Dequeue Operation – Algorithm to delete an element from a queue

Step 1: IF FRONT = _____ OR FRONT > _____

Write UNDERFLOW

ELSE

SET VAL = _____

SET FRONT = _____

[END OF IF]

Step 2: EXIT

Answer: Step 1: IF FRONT = -1 OR FRONT > REAR

Write UNDERFLOW

ELSE

SET VAL = QUEUE[FRONT]

SET FRONT = FRONT + 1

Step 2: EXIT